Mid-Point Summary

Phases A–B: landscape, competitive intelligence, telemetry analytics, and anomaly detection

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KEY DECISION TODAY

Select the default anomaly-detection path for Rover / Sentinel-style fleet health monitoring.

10
seeded train/test splits

150
controlled fault events

0.985
overall LOF AUROC

Phase A–B review frame

A four-week path from ecosystem orientation to statistically tested anomaly recommendation

Week	Focus	Deliverables
Week 1	Landscape + toolchain	Physical AI brief, PIC 2.0 map, Python environment
Week 2	Competitive intelligence	Vendor matrix, three strategic gaps, InGen crosswalk
Week 3	Telemetry analytics	Rover generator, EDA, feature dictionary
Week 4	Anomaly benchmark	Fault injection, 4-method test, recommendation memo

WHAT HAS BEEN DELIVERED

Foundation 5-reading landscape synthesis; PIC 2.0 conceptual map; reproducible Python 3.11 toolchain.

Data system Seedable Rover telemetry generator plus EDA and 25-feature dictionary.

Market view Competitive matrix across humanoid, service, and patrol vendors; gap memo tied to InGen platforms.

Evidence base Controlled fault injection; 4-method benchmark; Wilcoxon-tested recommendation.

Mid-point question: which anomaly-detection method should become the default path for Phases C–D integration?

Phase A: strategic grounding

Weeks 1–2 established the product context and competitor baseline for the technical benchmark

CORE CONCLUSION

Physical AI is moving toward a platform contest, not a single-robot contest.

That shifted the analysis from “which robot is best?” to “which intelligence, telemetry, safety, and learning layers can compound across products?”

THREE STRATEGIC GAPS CARRIED INTO PHASE B

1	**Fragmented competitor field**	Most vendors emphasize one body or one vertical; InGen can frame “one intelligence layer, many embodiments.”
2	**Under-served human-centered adoption**	Care, education, and companion use cases need trust, safety, and decision quality, not only robot hardware.
3	**Security needs governance-aware anomaly logic**	Patrol/security systems require uncertainty, escalation discipline, and auditable false-alert control.

PRODUCT ANCHOR LOGIC

Origami / PIC 2.0

Foundation layer for uncertainty, safety, adaptive control, and fleet learning.

Aido Rover + Sentinel

Best Phase B substrate: telemetry-rich, operations-facing, and naturally tied to anomaly detection.

Implication: use Week 4 to test whether fleet-health monitoring can be made statistically defensible, not just plausible.

Week 3: telemetry foundation

A reproducible Rover dataset and feature layer created the substrate for the Week 4 benchmark

432K
synthetic rows in default EDA sample

5
Rover units in sample

129.6M
generator full-scale capacity

25
documented features

1 Hz
sample rate

Generator coverage

- joint / actuator angles
- IMU accel + gyro
- motor current per drive
- battery voltage + SoC
- WiFi RSSI
- GPS fix quality
- task-success flags

Feature engineering logic

Short-window statistics	rolling means, slopes, RMS vibration, missingness rate
Cross-sensor ratios	voltage vs SoC, speed vs current, GPS vs WiFi consistency
Fleet-relative z-scores	battery, motor current, RSSI compared across units

Why it matters for Week 4

The benchmark compares detectors on engineered signals that carry physical meaning, not only raw sensor values.

Week 4 benchmark design

Controlled fault injection with leakage controls and paired statistical testing

FAULT PATTERNS INJECTED INTO TEST BLOCKS ONLY

Motor stall — speed collapse + one drive current spike + thermal rise

GPS jitter — spoofing-like position jumps + HDOP instability

Sensor drift — gradual IMU accel / gyro calibration shift

Software hang — stale telemetry blocks + task-success impact

Battery degradation — accelerated SoC drain + voltage sag

EVALUATION DISCIPLINE

10 seeded splits — same split seeds across all four methods

Clean train windows — faults injected after split and only in held-out blocks

Scaler fit on train only — no test-window statistics leak into normalization

Paired Wilcoxon tests — AUROC differences compared split-by-split

Design choice:
evaluate one-minute windows rather than isolated seconds, reflecting how an operations dashboard would triage fleet health.

Benchmark results: LOF leads accuracy

Overall metrics across five fault types and 10 seeded train/test splits

Overall AUROC and F1

Accuracy evidence

LOF beat Isolation Forest by +0.141 AUROC and LSTM AE by +0.137 AUROC; both paired Wilcoxon $p = 0.002$. LOF also beat One-Class SVM by +0.003 AUROC with $p = 0.0195$, but the operational gap is small.

Per-fault readout

LOF ranked best on 4/5 faults. One-Class SVM was marginally best on software hang and effectively tied with LOF.

Recommendation by operating regime

The preferred method changes depending on latency, accuracy, and escalation risk

Latency-constrained edge screen

Use Isolation Forest as a first-pass screen only, paired with explicit guards for battery, GPS, stale telemetry, and task-success degradation.

Caveat: benchmark AUROC was 0.844 ± 0.044, materially below LOF and One-Class SVM.

Accuracy-maximizing analytics

Use LOF as the default model for offline review, fleet-health triage, and analyst dashboards.

Caveat: neighbor-based scoring may become heavier at 50-unit, 30-day scale and needs stress testing.

High-stakes escalation

Use LOF plus rule-based safety checks; keep One-Class SVM as the challenger for software-hang-like faults.

Caveat: thresholds must be calibrated to severity, false-alert tolerance, and operator workload.

Explicit limitations

- Synthetic telemetry and controlled faults, not real field failures.
- One-minute windows do not measure sub-second alert latency.
- Fixed contamination threshold should be recalibrated before field use.
- Re-run needed on the full 50-unit, 30-day sample before final deployment claims.

Bottom line:
adopt LOF as the primary accuracy path, keep One-Class SVM as challenger, and use Isolation Forest only as a lightweight screen with guardrails.

Next phase: convert analysis into policy and dashboard decisions

Phases C–D will test decision behavior and package the results for stakeholders

What is ready now

- Clean repo structure and Week 1–4 artifacts
- A seedable Rover telemetry generator
- EDA and feature dictionary aligned to fleet health
- Benchmark pipeline with leakage controls and significance testing
- Recommendation memo with operating-regime caveats

Phases C–D focus

Week 5	PPO / SAC benchmarks and multi-agent coordination lessons
Week 6	AMDC-style decision-evaluation harness with reliability analysis
Week 7	Streamlit dashboard: fleet health, policy benchmark, decision scorecards
Week 8	Capstone report and executive deck

Mid-point message:
Phases A–B produced a defensible evidence chain from landscape framing to statistically tested anomaly-method selection.